Siddhant Roy

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Education

- **Jaypee Institute of Information Technology, Noida Sector-128** — **Nov. 2022 - present**
  Bachelor of Technology in Electronics and Communication Engineering
- **Arwachin Bharti Bhawan Sr. Secondary School** — **Apr. 2019 - Mar. 2021**
  Senior Secondary, 87.6%
- **Arwachin Bharti Bhawan Sr. Secondary School** — **Apr. 2011 - Mar. 2019**
  Secondary, 87.8%

Skills

- Programming Languages: Python, C/C++
- Frameworks & Libraries: Tensorflow, OpenCV, GANs, YOLOv5
- Databases & Data Handling: MySQL , CSV based databases
- Version Control & Tools: Git, GitHub

Projects

Automatic Billing System | AI, YOLOv5, Computer Vision

- Engineered an AI-driven automated billing system utilizing YOLOv5 for real-time object detection and price calculation.
- Accurately identifies retail items through image recognition, retrieves prices from a CSV-based database, and generates a detailed bill without manual barcode scanning.
- Optimized the detection model to handle similar packaging designs, improving recognition accuracy and reliability.

Deepfake Generation and Detection | GANs, AI Security

- Developed a deepfake generation and detection system by building a custom GAN (Generative Adversarial Network) from scratch, instead of using pre-existing models like StyleGAN.
- Designed the model to generate realistic synthetic faces using a noise vector (Z vector) as input.
- Implemented a deepfake detection system to identify AI-generated forgeries, enhancing media authenticity and security.

Positions of Responsibility

- **Head, Fortissimo – The Music Society of JIIT 128** — **Aug. 2024 - present**
  - Led a team of 50+ members in organizing musical events, concerts, and competitions.
  - Managed event planning, budgeting, and coordination with college authorities.
  - Conducted music workshops and jam sessions to mentor aspiring musicians.
  - Organized and performed in inter-college and national-level music competitions.
  - Under my leadership, the society secured 1st and 2nd positions in multiple inter-college Battle of the Bands events.
  - Expanded society membership and engagement through social media and campus outreach.
- **Music Head, Abhivyakti - Dramatics Society of JIIT 128** — **Aug. 2024 - present**
  - Managed the music team, guiding them through rehearsals and performances.
  - Created background music for Nukkad (street play) scenes to enhance their impact.
  - Worked closely with the drama team to match music with the emotions and themes of the play.
  - Performed live music during plays at various competitions and fests.
  - Helped grow the society's presence through creative and engaging music.